CCV: Compositional Consensus Verification for Multimodal Geometry Reasoning

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Abstract

Multimodal geometry reasoning requires models to jointly perceive diagrams and perform symbolic derivations. Test-time verification methods such as majority voting assume voter errors are roughly independent, but in geometry this assumption breaks down: all candidates perceive the same diagram, so perceptual biases cascade into correlated errors that produce confident but wrong consensus. We introduce SKETCH, a format that decomposes each candidate into a neural pathway (the model's stated answer, or claim) and a symbolic pathway (visual grounding $\rightarrow$ typed DSL $\rightarrow$ deterministic execution). The two pathways are decoded in the same rollout and are positively correlated, yet their agreement is strongly diagnostic: on our main model, a candidate's claim matches its execution about four times as often when the execution is correct as when it is wrong. Our method, Compositional Consensus Verification (CCV), gates consensus on this cross-channel agreement, so each concordant vote is far more precise than a raw majority vote. Given a pool of SKETCH candidates from a LoRA-tuned generator, CCV needs no external scorer and no additional model calls. Across four geometry evaluation sets, CCV reaches 81.0% accuracy, +10.4 pp over majority voting. Seven non-compositional baselines, including trained PRMs, cluster at 70–71%, while compositional cross-layer signals reach 79–81%.

1 Introduction

The verification bottleneck in geometry reasoning. Geometry reasoning differs from purely textual mathematical tasks in a fundamental way: it requires joint visual perception and symbolic derivation. A model must read spatial relationships from a diagram—angles, parallel lines, congruences—and then chain these percepts through multi-step logical deductions. This conjunction makes geometry a task that current vision-language models find difficult [Gao et al., 2023, Xia et al., 2024, Zhang et al., 2024b], and it also creates a distinctive failure mode for test-time verification. Figure 1 summarizes our proposed solution.

The standard approach for improving reasoning accuracy at test time is self-consistency [Wang et al., 2022]: sample K candidates and select the most frequent answer via majority voting (MV). This rests on the Condorcet assumption that voter errors are approximately independent, so that the majority converges to truth as K grows. In geometry, this assumption breaks down in a specific and measurable way. All K candidates perceive the *same* diagram; if the model harbors a systematic

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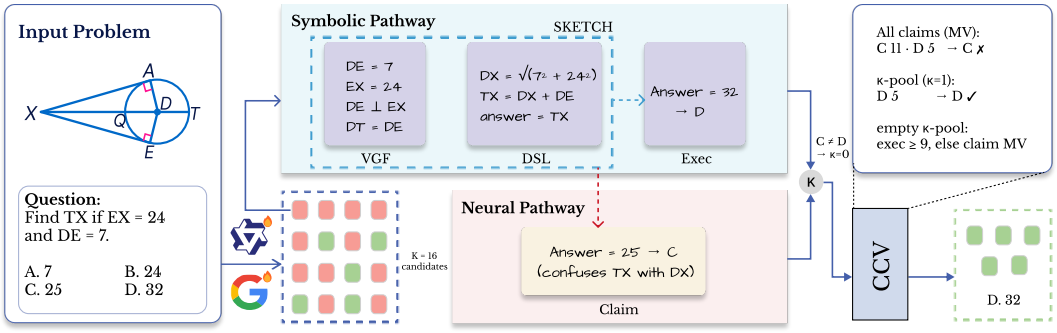


Figure 1: **Method overview.** Each of the K=16 candidates writes a SKETCH (visual grounding facts and a typed DSL program) and then states its answer (claim); an external interpreter executes the DSL (exec). The two pathways share the SKETCH and diverge only at the last step. In the expanded candidate, the claim (C) disagrees with the execution result (32 → D), so κ=0. Majority voting over all claims selects C (11/16) and is wrong; CCV keeps only the five κ-consistent candidates, which all answer D. If no candidate is κ-consistent, CCV falls back to an execution majority of at least τ_{exec}=9 votes, and otherwise to the claim majority.

perceptual bias—misidentifying a perpendicular as supplementary, or misreading an angle label—then a majority of candidates will share the same error. Votes become correlated because they reflect the same perceptual errors. Formally, when the pairwise error correlation ρ is substantial, the effective sample size degrades as $N_{\text{eff}} = K/(1 + (K-1)\rho)$ [Ladha, 1992, Nitarach, 2026]. At $\rho = 0.5$, sixteen candidates carry the information of barely two independent votes. In our experiments, wrong execution consensus accounts for 86% of the recoverable failures of execution-based voting (Section 4.6): confident majorities that derive from the same misperceived geometric facts.

Compositional structure as a verification resource. Since the votes share their errors, more votes do not help. What a verifier needs instead is a per-candidate signal that occurs at different rates for correct and incorrect candidates. Neuro-symbolic candidates carry such a signal because they are not opaque votes. Unlike standard chain-of-thought outputs—a single text sequence mapping to a single answer—a neuro-symbolic candidate decomposes into layers with distinct computational character:

$$\text{Candidate} = \big(\underbrace{\text{VGF}}_{\text{perception}} , \underbrace{\text{DSL}}_{\text{logic}}, \underbrace{\text{exec}}_{\text{computation}} , \underbrace{\text{claim}}_{\text{answer}} \big) \tag{1}$$

where VGF (Visual Grounding Facts) captures geometric relations extracted from the diagram, DSL (Domain-Specific Language) is a typed program encoding the reasoning steps, exec is its deterministic execution result, and claim is the model's stated answer.

The claim and the execution result are produced by different mechanisms. The model decodes each candidate in one autoregressive pass, in the order VGF → DSL → claim; 27,342 of the 27,344 Qwen3.5-9B completions follow this order. The claim is therefore conditioned on the DSL source but not on its output, because execution runs afterwards in an external sandboxed interpreter. The two pathways share their upstream and diverge at the last step: the model states an answer, while the interpreter computes one from the model's program. They are positively coupled. Across the Qwen3.5-9B pool, $P(\text{exec correct} \mid \text{claim correct}) = 0.736$, against a marginal execution accuracy of 0.617. Within a question, the correlation between claim and execution correctness averages $\phi = 0.325$ over the 766 questions where both vary (failed executions count as incorrect), compared with $\rho_e = 0.526$ between the executions of two candidates.

Existing verification paradigms do not use this structure. *Consensus methods* (MV, self-consistency, Confidence-Informed Self-Consistency (CISC) [Taubenfeld et al., 2025]) count votes within a single output layer, making them vulnerable to within-layer error correlation. *Learned verifiers* (Outcome Reward Models (ORM) [Cobbe et al., 2021], Process Reward Models (PRM) [Lightman et al., 2024, Wang et al., 2024], VisualPRM [Wang et al., 2025]) train an external model to score candidates, which requires domain-specific training data and adds inference cost. Neither paradigm exploits the compositional structure that is already present in the candidates. We propose a third approach: **compositional verification**, which derives verification signal from cross-pathway agreement within the candidates themselves.

Agreement as a likelihood-ratio signal. The two pathways need not be independent for their agreement to be informative. Filtering candidates by claim–execution agreement raises per-candidate precision over raw execution voting if and only if agreement is more likely when the execution is correct than when it is wrong (Proposition 1). The condition assumes nothing about the generation process and can be measured on any labeled candidate pool. Agreement is $4.18\times$ more likely for correct executions on Qwen3.5-9B and $2.96\times$ on Gemma4-E4B. False agreement need not be rare: on Gemma4-E4B, 40.2% of agreeing candidates are wrong, yet their precision (59.8%) is far above that of raw execution votes (33.4%).

Contributions. This paper turns the internal structure of neuro-symbolic candidates into a test-time verification signal. We introduce the SKETCH format, which decomposes geometry reasoning into VGF, typed DSL, deterministic execution, and a stated claim while maintaining >99.9% format compliance with a 9B VLM. We give an exact, measurable condition under which claim–execution agreement improves per-candidate precision, without assuming independence between the pathways. Building on it, we propose CCV, a training-free verifier that gates scoring and output routing through claim–execution agreement. On an existing SKETCH pool it makes no additional model calls and improves over majority voting by $+10.41$ pp across four evaluation sets. The same fixed protocol transfers to Gemma4-E4B ($+7.84$ pp over Claim MV). Question by question, CCV fixes 237 majority-voting errors while introducing 59 new ones.

2 The SKETCH Format

Visual grounding facts (VGF). For cross-candidate verification to work, visual observations must be comparable across candidates in a structured way. Free-form natural-language descriptions of a diagram are not reliably alignable; two candidates may describe the same geometric relationship with different wording. We therefore represent visual grounding as a list of atomic facts drawn from a closed vocabulary of relations:

$$\begin{gathered} f = (\texttt{relation}, \texttt{entities}), \\ \texttt{relation} \in \{\texttt{parallel}, \texttt{perpendicular}, \texttt{collinear}, \ldots\} \end{gathered} \tag{2}$$

The closed vocabulary ensures parseability and enables direct cross-candidate comparison of perceived geometric structure.

Typed DSL. The DSL is a restricted Python subset that derives the answer from VGF-grounded premises using geometry helper functions and standard mathematical operations ($\sqrt{\cdot}$, trigonometric functions, π, etc.). Restricting the subset—no imports, no I/O, no loops—ensures termination and deterministic execution, which is essential for the symbolic pathway to produce a reliable verification signal. Every program terminates with an explicit `answer = ...` assignment, so the execution result can be extracted automatically without answer parsing.

Closed-loop structure. The key property of the SKETCH format is that each candidate carries an intrinsic verification signal: whether its neural claim agrees with its symbolic execution. This signal exists because the format closes the loop: VGF provides premises, the DSL reasons from them, the model states its answer, and the interpreter executes the DSL to produce a result that can be compared with that answer. When claim $\approx$ exec (compositional consistency, $\kappa{=}1$), the model's stated answer coincides with what its own program computes; Section 3 shows when this agreement is evidence of correctness.

Training. We fine-tune Qwen3.5-9B with LoRA ($r{=}8$, $\alpha{=}16$) on 2,012 geometry examples derived from the Geometry3K [Lu et al., 2021] training split. We use the geometry3k subset of NVIDIA Nemotron-Image-Training-v3 [Blakeman et al., 2025], which provides 2,078 CoT annotations generated by Qwen3-VL-235B. Each CoT trace is reformatted into VGF+DSL structure by DeepSeek V4 Pro. The resulting model achieves format compliance >99.9%, VGF parse success >98.7%, and DSL execution success >94.5% on test data.

3 Compositional Consensus Verification

3.1 Verification Signal Taxonomy

Given K candidates $\{c_1, \ldots, c_K\}$, each decomposed as $c_i = (\text{VGF}_i, \text{DSL}_i, \text{exec}_i, \text{claim}_i)$, we define a hierarchy of verification signals ordered by the structural diversity they require:

Level 1: Within-layer consensus. Signals computed within a single output layer, aggregated across candidates:

$$S_{\text{claim}}(z) = |\{i : \text{claim}_i = z\}| \qquad \text{(surface consensus)} \tag{3}$$

$$S_{\text{exec}}(z) = |\{i : \text{exec}_i = z,\ \text{exec}_i \neq \perp\}| \qquad \text{(execution consensus)} \tag{4}$$

Traditional MV selects $\arg\max_z S_{\text{claim}}(z)$. Execution MV selects $\arg\max_z S_{\text{exec}}(z)$.

Level 2: Cross-layer agreement. Signals that compare outputs across structurally different layers. The core signal is compositional consistency:

$$\kappa_i = \mathbf{1}[\text{claim}_i \approx \text{exec}_i \wedge \text{exec}_i \neq \perp], \tag{5}$$

which checks whether a single candidate's neural and symbolic pathways agree. Additional cross-layer features (e.g., cross-candidate cross-layer agreement μ_i) appear in the learned ablation (Section 4.5) but are not part of the hand-designed algorithm.

Level 3: Agreement-gated consensus. Cross-layer agreement gates within-layer consensus:

$$G_\kappa(z) = \sum_i \mathbf{1}[\text{exec}_i = z] \cdot \kappa_i. \tag{6}$$

Only execution votes from self-consistent candidates count. This is the signal that CCV's scoring function is built on.

3.2 The Cross-Channel Verification Advantage

For notational simplicity, this subsection writes claim–exec agreement as equality; in implementation, κ_i uses approximate numeric matching and excludes failed executions.

Theorem 1 (Cross-channel agreement as a precision filter)**.** *Let X_i be the claim answer and Y_i the executed answer for candidate i, and let z^* be the correct answer. Define $\kappa_i = \mathbf{1}[X_i = Y_i]$, $t_\kappa = P(X_i = Y_i = z^*)$, and $f_\kappa = P(X_i = Y_i \neq z^*)$. Then the accuracy of self-consistent candidates satisfies the identity*

$$p_\kappa \triangleq P(Y_i = z^* \mid \kappa_i = 1) = \frac{t_\kappa}{t_\kappa + f_\kappa}. \tag{7}$$

Proposition 1 (Likelihood-ratio condition)**.** *Let $E_i = \mathbf{1}[Y_i = z^*]$, counting failed executions as $E_i = 0$ and $\kappa_i = 0$, and let $p_e = P(E_i = 1) \in (0, 1)$. Write $a_1 = P(\kappa_i = 1 \mid E_i = 1)$ and $a_0 = P(\kappa_i = 1 \mid E_i = 0) > 0$. Then*

$$\frac{p_\kappa}{1 - p_\kappa} = \frac{a_1}{a_0} \cdot \frac{p_e}{1 - p_e}, \tag{8}$$

so $p_\kappa > p_e$ if and only if $a_1 > a_0$.

Theorem 1 and Proposition 1 isolate the mechanism: κ-gating is a precision filter. It needs neither repeated votes in one channel nor independence between channels; it needs agreement to occur at different rates for correct and incorrect executions. Table 1 reports the two rates. Plugging them into the proposition reproduces the measured precision of agreeing candidates: 0.871 on Qwen3.5-9B and 0.598 on Gemma4-E4B.

Theorem 2 (One-vs-rest SNR approximation)**.** *In a one-vs-rest binary correctness approximation, majority reliability depends on per-vote accuracy p, pool size n, and pairwise correctness correlation ρ through*

$$\text{SNR}(p, n, \rho) = \frac{p - \frac{1}{2}}{\sqrt{p(1-p)(1 + (n-1)\rho)/n}}. \tag{9}$$

The κ-filtered pool improves over raw exec voting whenever the precision gain from filtering outweighs the combined penalty of smaller pool size and higher correlation (full condition in Appendix L).

Table 1: Agreement is 3–4× more likely when execution is correct, on both models. Failed executions count as wrong executions. The within-question ratio (Mantel–Haenszel) excludes failed executions. Excluding failed executions entirely gives a Qwen3.5-9B ratio of 3.51×.

	Qwen3.5-9B	Gemma4-E4B
$P(\text{agree} \mid \text{exec correct})$	0.802	0.646
$P(\text{agree} \mid \text{exec wrong})$	0.192	0.218
Ratio	4.18×	2.96×
Within-question ratio (Mantel–Haenszel)	3.22×	2.09×
CCV gain over Claim MV	+10.41 pp	+7.84 pp

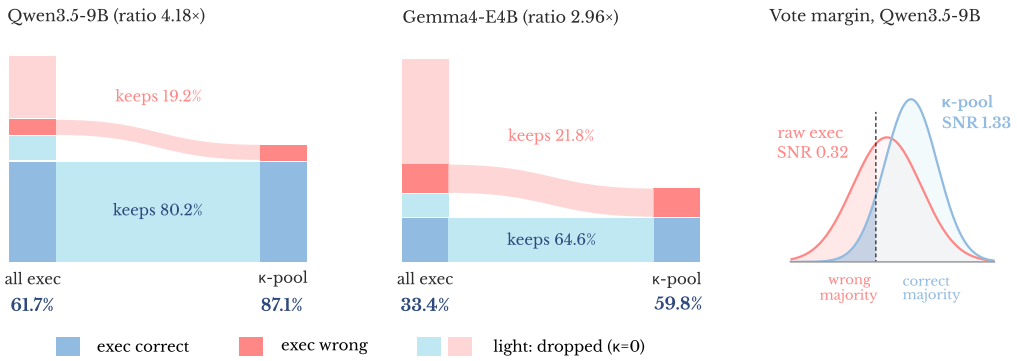


Figure 2: **Cross-channel precision filtering.** Left: κ-gating keeps 80.2% of correct executions but only 19.2% of wrong ones on Qwen3.5-9B (64.6% vs. 21.8% on Gemma4-E4B), raising the share of correct executions from 61.7% to 87.1% (33.4% to 59.8%). Light segments are dropped candidates (κ=0). Right: one-vs-rest Gaussian approximation of the vote margin on Qwen3.5-9B; the dashed line marks the majority threshold. The κ-pool is smaller (n: 16 → 9.09) and more correlated (ρ: 0.526 → 0.646), but its higher precision raises the SNR from 0.32 to 1.33.

This yields a testable, counterintuitive prediction: κ-gating need not reduce correlation to improve majority reliability. Precision filtering can dominate the correlation penalty. This complements the negative findings of Nitarach [2026] on diverse prompting: when errors arise from shared perception, decorrelating samples is the wrong lever—filtering for cross-channel agreement targets precision directly.

Proposition 1 governs per-candidate precision exactly. Whether the vote itself improves also depends on the retained pool's size K_κ and correlation ρ_κ, which Theorem 2 captures only approximately, so the precision condition alone does not guarantee a pool-level gain.

Empirical support. We directly measure within-question correctness correlations on the full pool (1,709 evaluation instances × 16 candidates). Raw exec votes have pairwise correlation $\rho_e = 0.526$; κ-filtered votes have higher correlation $\rho_\kappa = 0.646$. Yet per-candidate accuracy jumps from 61.7% to 87.1%, and the one-vs-rest SNR proxy improves by $4.1\times$ (0.32 to 1.33). Figure 2 visualizes this precision-filtering regime. Thus the retained pool is more correlated, but substantially more reliable—directly validating the counterintuitive prediction of Theorem 2: precision filtering dominates even when it increases correlation. This confirms that the dominant mechanism is precision filtering, not decorrelation: the $\kappa_i \cdot |\text{exec_cluster}_i|$ term in Algorithm 1 counts consensus only after it has survived cross-channel agreement.

3.3 κ-gated Dual-Pathway CCV

Algorithm 1 summarizes CCV. It uses claim-execution agreement as a precision gate, conceptually related to program-based consistency filtering (PROVE, PROgrams as VErifiers) [Toh et al., 2024], but differs by combining agreement-gated consensus with output-pathway routing. The algorithm applies cross-channel agreement as a gate at two levels:

Algorithm 1 κ-gated Dual-Pathway CCV

Require: Candidates $\{c_1, \ldots, c_K\}$, threshold τ_{exec}

1: **for** each candidate c_i **do**
2: $\quad \kappa_i \leftarrow \mathbf{1}[\text{claim}_i \approx \text{exec}_i \wedge \text{exec}_i \neq \perp]$
3: $\quad \text{score}_i \leftarrow 0.5 \cdot |\text{claim_cluster}_i| + \kappa_i \cdot (1 + |\text{exec_cluster}_i|)$
4: **end for**
5: $\mathcal{C}_\kappa \leftarrow \{c_i : \kappa_i = 1\}$ ▷ self-consistent candidate pool
6: **if** $\mathcal{C}_\kappa \neq \emptyset$ **then**
7: $\quad$ **return** $\text{claim}(\arg\max_{c_i \in \mathcal{C}_\kappa} \text{score}_i)$ ▷ *Level 1: output neural claim*
8: **else**
9: $\quad z_{\text{exec}} \leftarrow \text{mode}(\{\text{exec}_i : \text{exec}_i \neq \perp\})$
10: $\quad$ **if** $|\{i : \text{exec}_i = z_{\text{exec}}\}| \geq \tau_{\text{exec}}$ **then**
11: $\quad\quad$ **return** z_{exec} ▷ *Level 2: fall back to symbolic execution*
12: $\quad$ **else**
13: $\quad\quad$ **return** $\text{mode}(\{\text{claim}_i\})$ ▷ *Level 3: fall back to claim majority*
14: $\quad$ **end if**
15: **end if**

Candidate scoring (Lines 2–3). The term $\kappa_i \cdot |\text{exec_cluster}_i|$ realizes agreement-gated consensus: execution consensus is trusted only when accompanied by cross-layer agreement. The term $0.5 \cdot |\text{claim_cluster}|$ provides a baseline surface signal. Raw execution cluster size receives zero weight—our learned ablation in Section 4.5 confirms this design choice.

Output pathway routing (Lines 6–13). When self-consistent candidates exist ($\mathcal{C}_\kappa$ is non-empty), the algorithm outputs the neural claim. When cross-layer agreement collapses globally, it falls back to symbolic execution consensus, but only if the execution cluster is sufficiently large ($\geq \tau_{\text{exec}}$). This routing treats the *absence* of cross-channel agreement as diagnostic: it signals that the neural pathway is unreliable, making the deterministic symbolic pathway more trustworthy.

Complexity. $O(K)$ time, zero additional model calls, one tunable parameter (τ_{exec}).

Cost. Verification is post-hoc aggregation over generated candidates and takes about 2–4 ms per question on one CPU core. Executing all 27,344 DSL programs takes under 10 s, a cost shared with Exec MV, Execution-Answer Voting (EAV) and the PROVE-style filter; SKETCH completions average about 86 tokens. By contrast, URSA-RM (URSA Reward Model) [Luo et al., 2025] weighted voting needs one reward-model forward pass per candidate (27,344 in our evaluation), and CISC one self-judgment pass per candidate. CCV is not a drop-in verifier for arbitrary candidate pools: it requires a generator fine-tuned to emit SKETCH (LoRA SFT (Supervised Fine-Tuning) on 2,012 examples, about 3.5 aggregate GPU-hours to the evaluated checkpoint and 10.6 for the full run) and the offline reformatting pipeline of Appendix A.

4 Experiments

4.1 Setup

Models and training. Our primary generator is Qwen3.5-9B, fine-tuned with LoRA (r=8, α=16) on 2,012 executor-validated Geometry3K training examples. The training data was constructed by reformatting Qwen3-VL CoT traces into the SKETCH format using DeepSeek V4 Pro. We also evaluate Gemma4-E4B to test cross-model generalization. Training was conducted using LLaMA-Factory with bf16 on 8 NVIDIA 4090 GPUs.

Evaluation protocol. Table 2 summarizes our evaluation suite. We use a frozen-rollout protocol with K=16 candidates, temperature 0.8, and top-p 0.95. All verification methods operate on the same rollout pool. The single hyperparameter τ_{exec}=9 was selected on a 300-question development split of Geometry3K and held fixed across all evaluation sets and models. Code, evaluation and aggregation scripts, the rollout pools of both models, per-question selections, external-scorer outputs and an audit script that recomputes every reported number are available at https://github.com/closss/neurips2026-paper.

Table 2: Test evaluation suite. We evaluate on one in-distribution set and three held-out/OOD settings. MathVista-GPS clean (146) is a subset of MathVista-GPS (208) and is reported separately; the All column pools the 1,709 evaluation instances, which cover 1,563 distinct questions.

Benchmark	Lang.	N	Role
Geometry3K test [Lu et al., 2021]	EN	601	In-distribution
GeoQA test [Chen et al., 2021]	ZH	754	Cross-lingual OOD
MathVista-GPS clean [Lu et al., 2023]	EN	146	Cross-domain OOD
MathVista-GPS [Lu et al., 2023]	EN	208	Reference

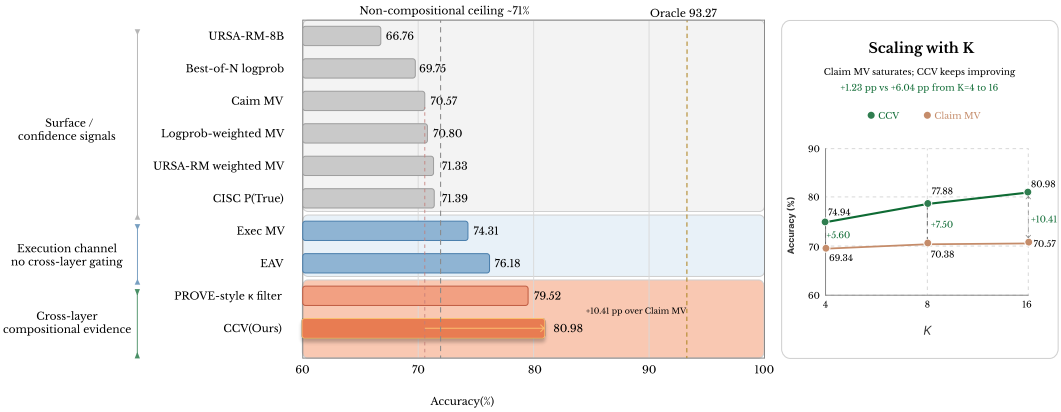


Figure 3: **Evidence ladder and K scaling.** Left: accuracy over the 1,709 evaluation instances. Surface and confidence-based selectors cluster near 70–71%, execution-only methods reach 74–76%, and cross-layer compositional verification reaches 79–81%; the dashed line marks the claim oracle (93.27%). Right: as K grows from 4 to 16, the CCV gain over Claim MV grows from $+5.60$ to $+10.41$ pp, while Claim MV itself gains only $+1.23$ pp.

4.2 Main Results

Figure 3 visualizes the same evidence ladder and K-scaling trend, while Table 3 reports the full evaluation-set breakdown. The four surface and confidence-based selectors—Claim MV, logprob-weighted MV, CISC, and URSA-RM weighted MV—all lie in a narrow 70.57–71.39% band, despite spanning self-confidence weighting and an external learned PRM. The learned ablation in Section 4.5 confirms this ceiling extends to logprob-based best-of-N and learned surface+execution feature selectors. Even URSA-RM, the strongest external scorer, achieves only $+0.76$ pp over Claim MV when used for weighted voting. Execution-only methods improve to 74–76%, but the large jump appears only when cross-layer agreement is used as a compositional signal: a PROVE-style κ filter reaches 79.52%, and CCV reaches 80.98%, a $+10.41$ pp improvement over Claim MV and $+4.80$ pp over EAV.

Paired per-question tests over the 1,709 evaluation instances confirm that these main gains are statistically reliable: CCV improves over Claim MV by $+10.41$ pp (95% paired bootstrap CI $[8.48, 12.40]$ pp; exact McNemar $p < 10^{-12}$) and over EAV by $+4.80$ pp (95% CI $[3.39, 6.26]$ pp; exact McNemar $p = 5.7\times10^{-11}$). The gain is also not concentrated on the development split: on the disjoint heldout/OOD split (excluding dev 300), CCV reaches 81.93%. Finally, a sweep over the only tunable parameter, τ_{exec}, shows that CCV is robust rather than finely tuned: across $\tau_{\text{exec}} = 5 \ldots 14$, dev accuracy varies by 1.00 pp and heldout/OOD accuracy by 1.16 pp (Appendix C). The κ matching threshold itself is also stable: varying the candidate-level relative tolerance around the default leaves CCV accuracy unchanged on the Qwen3.5-9B rollout pool (Appendix I).

4.3 Cross-Model Generalization

To test whether CCV's gains depend on the specific base model, we repeat the full evaluation protocol with Gemma4-E4B, a 4B-parameter multimodal model with substantially weaker geometry reasoning capacity. The Gemma4 model achieves comparable SKETCH format compliance to the Qwen model,

Table 3: Evidence ladder across verification signal types (K=16). Surface/confidence methods, including learned PRM scores and self-judged confidence, cluster near 70–71%. Execution-only signals improve moderately. Cross-layer compositional verification produces the large jump to 79–81%. Purple shading and bold mark the best non-oracle accuracy in each column. G3K = Geometry3K, GPS-c = MathVista-GPS clean. EAV (Execution-Answer Voting) outputs the largest execution cluster when it has at least 5 votes, and otherwise selects by a weighted score of claim-cluster size, execution-cluster size and κ. Claim oracle counts a question as solved if any of the 16 candidates states the correct answer. MathVista-GPS clean (146) is a subset of MathVista-GPS (208) and is reported separately; the All column pools the 1,709 evaluation instances, which cover 1,563 distinct questions.

Method	Type	G3K	GeoQA	GPS-c	GPS	All
Surface / confidence signals						
Claim MV	consensus	66.72	73.61	72.60	69.23	70.57
Logprob-weighted MV	weighted consensus	66.56	73.87	72.60	70.67	70.80
CISC $P(\text{True})$-weighted	confidence consensus	68.05	74.27	72.60	69.71	71.39
URSA-RM weighted MV	learned verifier	62.73	75.60	79.45	75.00	71.33
Execution channel without cross-layer gating						
Exec MV	exec consensus	72.71	75.60	76.71	72.60	74.31
EAV	exec-first priority	74.04	77.72	77.40	75.96	76.18
Cross-layer compositional signals						
PROVE-style κ filter	binary agreement	76.54	**82.10**	81.51	77.40	79.52
CCV (ours)	gated + routing	**78.54**	81.83	**86.30**	**81.25**	**80.98**
Claim oracle	any correct claim	93.51	93.50	91.78	92.79	93.27

so this experiment is a confirmation of the verification mechanism rather than a new generator contribution.

Table F.1 reports the cross-model evidence ladder using the same τ_{exec}=9 for both models. First, the absolute accuracy gap is large: Claim MV drops from 70.57% on Qwen3.5-9B to 44.94% on Gemma4-E4B, consistent with the capacity difference. Second, the CCV gain pattern is preserved: CCV improves over Claim MV by $+10.41$ pp on Qwen and $+7.84$ pp on Gemma4. The Gemma4 result is especially diagnostic because its execution pathway alone is worse than the claim pathway (Exec MV 41.08% vs Claim MV 44.94%), making EAV ineffective (44.88%). Yet CCV still reaches 52.78%, showing that it exploits structural cross-channel agreement rather than the raw accuracy advantage of execution votes.

The mechanism also replicates. The likelihood-ratio condition holds on both models ($4.18\times$ and $2.96\times$; Table 1), κ-filtering yields nearly identical precision gains ($+25.40$ pp vs. $+26.38$ pp), and the correlation pattern repeats ($\rho_\kappa > \rho_e$ while SNR improves; Table D.1). The CCV advantage also scales with K on both models (Appendix D, Appendix G).

4.4 K Scaling Analysis

Figure 3 shows that the CCV advantage scales with K: the gain over Claim MV grows from $+5.60$ pp at K=4 to $+10.41$ pp at K=16 (tabulated in Appendix D). Surface consensus saturates quickly (Claim MV gains only $+1.23$ pp from K=4 to 16), while compositional verification continues extracting signal from larger candidate pools.

As a negative control, VGF consensus—another within-layer signal—not only fails to help but *hurts* performance when added: adding VGF consensus to an earlier scoring variant of CCV lowers accuracy from 80.05% to 79.93%. This reinforces the central finding: within-layer consensus—whether the layer is claim, execution, or VGF—is vulnerable to correlated perception errors. Only cross-layer gating provides robust verification signal.

4.5 Learned Feature Ablation

To validate that the compositional features carry the transferable verification signal—not just that our specific CCV rule is effective—we train candidate-level logistic regression classifiers on Geometry3K dev (300 questions) and evaluate on disjoint OOD evaluation sets.

Table D.3 reports the full ablation. The key result is that compositional features are the only transferable signal: logprob best-of-N, surface features, and surface+execution features remain near Claim MV (71.11–71.86% OOD), while adding cross-layer features jumps to 79.85% OOD and C-only features reach 81.02%. The learned model rediscovers the gating principle: raw exec_cluster_size has a negative coefficient (-2.40), while $\kappa \times$ exec_cluster is strongly positive ($+1.96$). A simple rule is therefore enough: PROVE-style κ filtering reaches 81.02% OOD, and dual-output CCV reaches 81.93% OOD.

4.6 Error Exchange and Recovery Analysis

Table 4: CCV fixes three to four baseline errors for every one it introduces.

Comparison	Both correct	Only CCV correct	Only baseline correct	Both wrong	Gain : loss
vs. Claim MV	1,147	237	59	266	4.0 : 1
vs. EAV	1,263	121	39	286	3.1 : 1

Table 4 compares CCV with Claim MV and EAV question by question. CCV fixes 237 Claim MV errors while introducing 59 new ones, and fixes 121 EAV errors while introducing 39. The net gain is positive on every evaluation set against both baselines; against Claim MV it ranges from $+8.22$ pp on GeoQA to $+13.70$ pp on GPS-clean.

We call an EAV failure recoverable when at least one candidate states the correct answer. Of the 355 recoverable failures, 305 (86%) come from a wrong execution consensus. We split these by the claim side, using clusters of at least 8 of 16 candidates: a correct claim cluster (Exec✗ Claim✓, 139 cases), a wrong claim cluster and no correct one (Exec✗ Claim✗, 132), or dispersed claims (Exec✗ Claim?, 34). CCV recovers 88 (63.3%), 14 (10.6%) and 12 of these, and 7 of the remaining 50 failures. In total it recovers 121 of the 355, exactly its gains over EAV (Figure E.1; Appendix E).

Regressions have two main sources. In 26 of the 59 regressions against Claim MV and 30 of the 39 against EAV, the selected κ-consistent group is wrong: both pathways converge on the same wrong answer, usually in small groups (median size 3 and 2). In the other 33 regressions against Claim MV, the κ-pool is empty and a wrong execution cluster of size at least τ_{exec} overrides a correct claim majority. In all 33, no candidate executes to the correct answer, and 15 are unanimous at 16/16. The same override accounts for 32 of the 51 unrecovered Exec✗ Claim✓ cases. This is the shared-perception failure diagnosed in Section 1, surfacing in the branch where the κ signal is absent. The remaining 9 regressions against EAV come from CCV's stricter fallback threshold ($\tau_{\text{exec}} = 9$ vs. 5; 7 cases) and from EAV's weighted-score fallback (2 cases).

Across all 27,344 candidates, κ-consistent candidates are correct 19% of the time when their execution cluster has size 1 and 93% at size 8 or more, so κ must be combined with gated consensus rather than used as a standalone binary filter.

5 Related Work

Geometry reasoning systems mainly improve the generator side of the problem. Specialized VLMs such as G-LLaVA [Gao et al., 2023], GeoX [Xia et al., 2024], and MAVIS [Zhang et al., 2024b] strengthen multimodal geometry solving on benchmarks such as Geometry3K [Lu et al., 2021] and GeoQA [Chen et al., 2021]. Neuro-symbolic solvers such as AlphaGeometry [Trinh et al., 2024, Chervonyi et al., 2025] and FormalGeo [Zhang et al., 2023] use symbolic structure to derive proofs or answers. Our work is complementary: it assumes a candidate pool already exists and studies how to verify candidates when shared diagram perception makes their errors correlated.

Test-time verification is usually based on either consensus or learned scoring. Self-consistency [Wang et al., 2022], universal self-consistency [Chen et al., 2023], and confidence-weighted CISC

[Taubenfeld et al., 2025] aggregate opaque answer votes, which works best when sampled errors are sufficiently independent. Learned verifiers, including ORMs/PRMs [Cobbe et al., 2021, Uesato et al., 2022, Lightman et al., 2024, Wang et al., 2024], GenRM [Zhang et al., 2024a], URSA-RM [Luo et al., 2025], and VisualPRM [Wang et al., 2025], train external scorers and add verifier-specific data or inference. CCV differs from both paradigms by deriving verification signal from structure already present inside each candidate, at the cost of a one-time adaptation of the generator to the SKETCH format.

The closest line of work uses programs or error-correlation analysis to diagnose voting failures. PROVE [Toh et al., 2024] filters candidates whose translated Python output matches the natural-language answer, a binary consistency signal analogous to κ. Our framework generalizes this idea to geometry-specific neuro-symbolic outputs, where VGF, typed DSL, deterministic execution, and neural claims form a hierarchy of cross-layer signals. Nitarach [2026] show that correlated errors limit majority voting; CCV targets this regime directly by gating consensus through cross-channel agreement rather than trying to decorrelate samples.

6 Limitations

Coupled pathways. Claim and execution come from one autoregressive pass, and the claim is decoded after the DSL source, so the two are positively coupled (within-question $\phi = 0.325$). Our analysis does not assume independence, but the likelihood ratio, and with it the gain, depends on how tightly a given generator ties its stated answer to its program.

Dependence on the SKETCH format. CCV needs candidates that expose parseable VGF, an executable DSL program and a stated answer. Our generators obtain this through LoRA SFT on teacher-reformatted traces (Section 2, Appendix A); the 9B model reaches $>$99.9% section compliance, $>$98.7% VGF parse success and $>$94.5% DSL execution success. When format compliance degrades, κ excludes failed executions, so the κ-pool shrinks and CCV falls back to execution or claim majority instead of failing silently. Whether frontier models that cannot be fine-tuned produce such candidates under few-shot prompting is untested.

Scope of the evaluation. All evaluation sets are geometry problems with numeric or multiple-choice answers, tested on two open-weight models. Other domains need a deterministic computation path and a stated answer over a comparable answer space; there, the likelihood ratio of Proposition 1 can be measured before any accuracy comparison. Open-ended proofs have no scalar answer to compare. A natural analogue compares a formal proof checker's verdict with the model's claim of what it proved, which redesigns the agreement criterion rather than tuning it.

Limits of selection. CCV selects among existing candidates and cannot create a correct one. The claim oracle is 93.27% on Qwen3.5-9B and 91.75% on Gemma4-E4B; the remaining questions require a better generator. Among recoverable failures, CCV rarely recovers cases where the claim majority is also wrong (10.6%), and when the κ-pool is empty, a large wrong execution cluster can override a correct claim majority (Section 4.6). These cases need evidence beyond claim and execution, such as independent verification of the extracted premises.

7 Conclusion

We have shown that neuro-symbolic reasoning candidates carry an underused verification resource: agreement between a candidate's stated answer and the execution of its own program. The two pathways are coupled, but agreement is several times more likely when execution is correct, and this likelihood ratio, not independence, is what makes agreement-gated consensus precise. CCV turns this into a training-free verifier over existing SKETCH pools. Non-compositional confidence, consensus and learned-score methods saturate around 70–71%, while κ-based compositional methods reach 79–81%. Whether the same holds beyond geometry is a hypothesis we have not tested. The likelihood-ratio condition makes it testable: in any domain whose candidates expose a deterministic computation and a stated answer over a comparable answer space, such as chart or table QA, physics diagrams or code generation, the ratio can be measured on a labeled candidate pool before any accuracy comparison, and a ratio near 1 predicts little precision gain.

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A Training Data Construction

Table A.1 summarizes the full construction pipeline for the 2,012-example SFT set. We start from the official Geometry3K [Lu et al., 2021] training split, use the geometry3k subset of Nemotron-Image-Training-v3 [Blakeman et al., 2025] as the Qwen3-VL-235B-A22B-Thinking CoT source, reformat each matched trace into the SKETCH VGF+DSL+answer format with DeepSeek V4 Pro, and retain only examples whose DSL executes and matches the official answer. The reformatting stage is text-only: the expensive visual reasoning is supplied by the Qwen3-VL CoT trace, while the official gold answer is used as the final target for machine-checkable reformatting.

Table A.1: Training data construction and filtering. Percentages are relative to the 2,078 matched CoT rows unless otherwise noted.

Stage	Count	Rate
Official Geometry3K train rows	2,101	–
Matched Qwen3-VL CoT traces	2,078	98.91% of official
Qwen3-VL boxed-answer correct	2,065	99.37%
DeepSeek V4 Pro reformat attempts	2,078	100.00%
Strict validator pass	1,963	94.47%
Recovered by tolerant VGF normalization	49	2.36%
Final executor-validated SFT rows	2,012	96.82%
Filtered: execution-answer mismatch	61	2.94%
Filtered: DSL execution/safety error	5	0.24%

The final 66 filtered examples consist of 61 execution-answer mismatches and five DSL execution/safety failures: one undefined `UB`, one undefined `z`, one undefined `m`, and two blocked-token errors. The tolerant VGF pass clears unsupported or over-arity visual facts only when the DSL and

answer remain executor-valid; a final post-processing pass keeps 2,012 rows, with 41 DSL snippets changed by the sanitizer and 6 VGF lists cleared.

B Data Construction Prompts

Stage 1: stored source prompt for frontier CoT traces. We did not rerun Qwen3-VL-235B-A22B-Thinking for this paper. We use an existing CoT source whose stored user messages follow the problem image plus the following prompt pattern:

```
Solve this geometry problem. Study the diagram and select
the correct answer.

{problem_text}
{choices}

Do not explain your answer but write only the answer label
(A, B, ...) in this format:
Answer: \boxed{...}.
```

Stage 2: VGF+DSL reformatting. DeepSeek V4 Pro receives no image input. It reformats the official problem, choices, official answer, and Qwen3-VL CoT into the target SKETCH format using the following template:

```
Convert the provided Qwen3-VL-235B natural-language geometry
chain-of-thought into the target VGF+DSL format.
Do not solve from scratch. Use the teacher CoT as the visual
and reasoning evidence, and use the official gold answer as
the final target.
Return exactly three sections in this order and no extra sections:
[VGF]
[DSL]
[Final Answer]

[VGF] rules:
Output a raw JSON list only. Do not use markdown fences.
Each item must be {"relation": <relation>,
"entities": [<tokens>]} with exact relation arity.
Allowed relations: {allowed_vgf_relations}.
Relation arities: {relation_arities}.
For unary shape relations, put the whole object in one entity
token, e.g.
{"relation":"parallelogram","entities":["MNPQ"]}.
Use at most 4 facts to stay safely under the validator limit
of 5 facts.
Use [] if no reliable visual fact is needed.

[DSL] rules:
Output executable Python assignment statements only.
Do not use markdown fences.
Do not include comments or prose.
Never write import statements, including 'from math import ...'.
Math functions are already available by name.
Every variable must be assigned before it is used.
Do not write placeholder equations with unknown symbols.
The final assignment must define a variable named answer.
The value of answer must evaluate to the official gold option
value; use the teacher CoT equations to derive that value.
Allowed math names: sqrt, pi, sin, cos, tan, radians, degrees,
asin, acos, atan, abs, min, max, round.
```

```
Do not read files, import modules, call external tools, or use
input/output functions.
[Final Answer] must be exactly the official gold option letter:
{gold}.
In [DSL], set answer to the numeric or symbolic value of
option {gold}:
{gold_value}.

Official problem:
{problem_text}
{choices}

Official gold answer: {gold}
Official gold option value: {gold_value}
Teacher boxed answer: {qwen_boxed_answer}

Teacher NL CoT:
{qwen_cot}
```

C τ_{exec} Sensitivity

The only tunable hyperparameter in CCV is the execution-majority threshold τ_{exec}, used only when the κ-pool is empty. We sweep $\tau_{\text{exec}} = 1 \ldots 16$ on the main Qwen3.5-9B K=16 rollout pool, using the same split protocol as the main paper: Geometry3K qid$<$ 300 for dev, and Geometry3K qid$\geq$ 300 plus GeoQA, MathVista-GPS clean, and MathVista-GPS for heldout/OOD evaluation. Figure C.1 plots the full sweep, and Table C.1 reports selected points. We use the resulting τ_{exec}=9 for both Qwen3.5-9B and Gemma4-E4B.

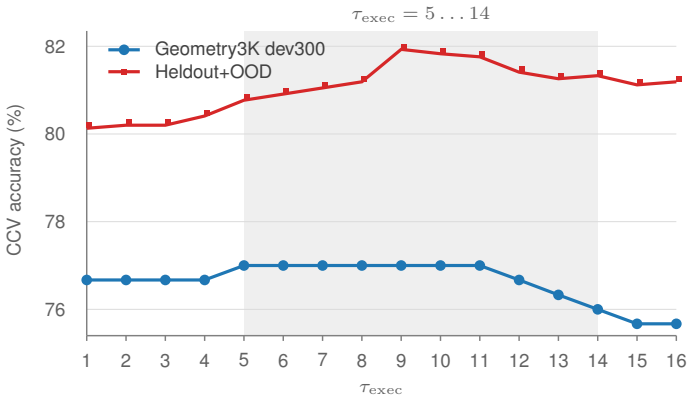


Figure C.1: **Sensitivity to τ_{exec}.** CCV is stable over a broad range of execution thresholds. In the plateau $\tau_{\text{exec}} = 5 \ldots 14$, dev accuracy varies from 76.00% to 77.00%, and heldout/OOD accuracy varies from 80.77% to 81.93%.

Table C.1: Selected points from the τ_{exec} sweep (full curve in the figure above). The dev-selected threshold is $\tau_{\text{exec}} = 9$.

τ_{exec}	Dev@300	Heldout/OOD	All
1 (no threshold)	76.67	80.13	79.52
9 (selected)	**77.00**	**81.93**	**80.98**
16 (maximum)	75.67	81.19	80.22

D Additional Diagnostics

This appendix contains the diagnostic tables referenced by the main experimental analysis. Table D.1 reports the precision-filtering statistics behind the cross-model claim, Table D.2 gives the exact Qwen K-scaling values summarized in Figure 3, and Table D.3 gives the learned feature ablation.

Table D.1: Cross-model diagnostic for the precision-filtering mechanism. The κ-filtered pool is more correlated than the raw execution pool, but its much higher per-candidate precision improves the effective verification signal.

Model	p_e (%)	p_κ (%)	Δp (pp)	ρ_e	ρ_κ	mean K_κ	SNR raw$\rightarrow\kappa$
Qwen3.5-9B	61.70	87.10	+25.40	0.526	0.646	9.09	0.32$\rightarrow$1.33
Gemma4-E4B	33.43	59.81	+26.38	0.504	0.630	5.77	−0.48$\rightarrow$0.24

Table D.2: Performance across K on the four-set Qwen3.5-9B pool.

K	Claim MV	Exec MV	CCV	Claim oracle	CCV − Claim MV
4	69.34	68.98	74.94	84.74	+5.60
8	70.38	72.43	77.88	89.89	+7.50
16	70.57	74.31	80.98	93.27	+10.41

Table D.3: Compositional feature ablation. "S" = standard features (logprob, claim cluster), "E" = execution features, "C" = compositional cross-layer features (κ, $\kappa \times$ exec_cluster, μ, $\kappa \times \mu$, sc_rate). The last column reports OOD gain over Claim MV.

Method / Features	Dev@300	OOD	All	Δ OOD vs Claim MV
Claim MV (reference)	66.67	71.77	70.57	0.00
Best-of-N (logprob)	66.00	71.11	69.75	−0.67
LogReg S-only	67.00	71.86	70.74	+0.08
LogReg S+E	66.33	71.69	70.63	−0.08
LogReg S+E+C	74.67	79.85	78.47	+8.08
LogReg C-only	75.33	81.02	79.64	+9.24
PROVE-style κ filter (rule)	75.00	81.02	79.52	+9.24
Dual-output CCV (ours, rule)	77.00	81.93	80.98	+10.16

E Error Flow and Recovery Taxonomy

Figure E.1 visualizes the error taxonomy and routing behavior. Table E.1 details the recovery breakdown across all 355 recoverable EAV failure cases.

Three details characterize these failure modes: First, among the 88 recoveries in the Exec✗Claim✓ category, 67 are routed through the κ-pool and 21 through the claim-majority fallback; among the 51 unrecovered cases, 32 are due to an empty κ-pool where a wrong execution cluster triggers execution fallback, and 19 stem from incorrect selection within the κ-pool. Second, among the 132 Exec✗Claim✗ cases, 69 have claim and execution majorities pointing to the same incorrect answer, while the remaining 63 point to different incorrect answers. Third, following Section 4.6, "recoverable" is evaluated solely on the claim channel: at least one candidate in the pool states the correct answer.

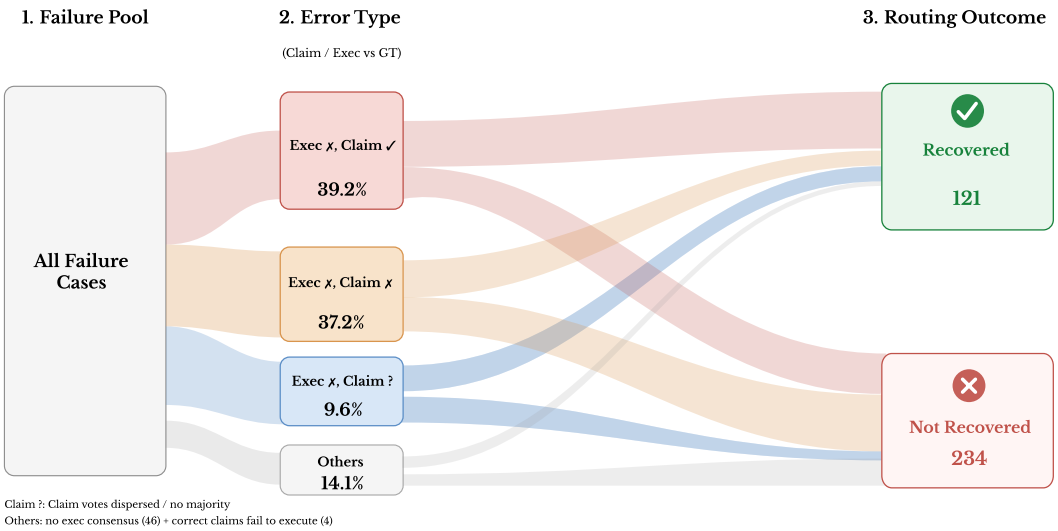


Figure E.1: **Error flow and recovery.** The 355 recoverable EAV failures split into a wrong execution consensus with a correct claim cluster of at least 8/16 (Exec✗ Claim✓, 139), a wrong claim cluster (Exec✗ Claim✗, 132) or dispersed claims (Exec✗ Claim?, 34), and 50 other failures. CCV recovers 88, 14, 12 and 7 of these: 121 recoveries in total, equal to its gains over EAV in Table 4.

Table E.1: CCV's 121 recoveries equal its gains over EAV. Recoverable failures are defined by the claim channel: at least one candidate states the correct answer.

Failure type (355 recoverable EAV failures)	Cases	Recovered by CCV	Rate
Wrong exec consensus, correct claim cluster (Exec✗ Claim✓)	139	88	63.3%
Wrong exec consensus, wrong claim cluster (Exec✗ Claim✗)	132	14	10.6%
Wrong exec consensus, dispersed claims (Exec✗ Claim?)	34	12	35.3%
No execution consensus selected by EAV	46	7	15.2%
Every correct-claim candidate fails to execute	4	0	0.0%
Total	355	121	34.1%

F Cross-Model Validation

Table F.1 evaluates the same verification protocol on a lower-capacity Gemma4-E4B generator, using the same τ_{exec}=9 as Qwen3.5-9B. The Gemma4 pool uses the same frozen K=16 four-set protocol as the main Qwen3.5-9B experiments: Geometry3K test (601), GeoQA test (754), MathVista-GPS clean (146), and MathVista-GPS (208), for N=1709 evaluation instances and 27,344 candidates. The evaluation uses the standard benchmark files; it does not use the deprecated 400-question checkpoint-selection set.

The notable difference is that Gemma4 reverses the raw pathway ordering: Exec MV is below Claim MV (41.08% vs 44.94%), whereas Qwen's execution pathway is stronger than its claim pathway. Consequently, EAV, an execution-first priority rule, is ineffective on Gemma4 (44.88% vs 44.94% Claim MV). This pattern is expected for a smaller model with weaker visual grounding: within-layer execution consensus is more easily dominated by correlated perception errors. CCV does not trust the raw execution vote; it uses claim–execution agreement as a precision filter. Even when the execution pathway is worse in aggregate, candidates where claim and execution agree remain informative, yielding $+7.84$ pp over Claim MV and $+7.90$ pp over EAV on Gemma4.

Table F.1: Cross-model validation on the same frozen K=16 four-set protocol (N=1709), using τ_{exec}=9 for both models. Absolute accuracy differs because Qwen3.5-9B has higher model capacity than Gemma4-E4B, but the gain pattern is consistent: cross-layer κ filtering and CCV improve over surface consensus for both models without additional model calls on an existing SKETCH pool.

Method	Qwen3.5-9B	Gemma4-E4B
Claim MV	70.57	44.94
Exec MV	74.31	41.08
EAV	76.18	44.88
PROVE-style κ filter	79.52	51.90
CCV (ours)	**80.98**	**52.78**
CCV − Claim MV	+10.41	+7.84
CCV − EAV	+4.80	+7.90
Claim oracle	93.27	91.75

G Gemma4 K Scaling

Table G.1 repeats the K-scaling analysis on the frozen Gemma4-E4B K=16 pool by subsampling candidates; K<16 rows are averaged over four random subsamples. We keep τ_{exec}=9 fixed to match the cross-model comparison in the main text.

Table G.1: Gemma4-E4B K scaling from the frozen K=16 pool. Entries for K=4, 8 are mean±std over four subsamples.

K	Claim MV	Exec MV	EAV	PROVE-style κ	CCV	Claim oracle	CCV − Claim MV
4	42.41±0.81	37.60±0.42	46.94±0.46	46.65±0.74	47.28±0.60	72.24±0.60	+4.87
8	44.00±0.48	39.86±0.24	48.82±0.79	50.04±0.62	50.85±0.58	83.84±0.32	+6.85
16	44.94	41.08	44.88	51.90	52.78	91.75	+7.84

The Gemma4 gain increases with candidate budget, from +4.87 pp at K=4 to +7.84 pp at K=16, matching the qualitative scaling trend observed for Qwen.

H Gemma4 Implementation Details

The Gemma4-E4B experiment uses the frozen checkpoint-720 LoRA adapter and the frozen rollout pool described in Appendix F; no additional model calls are made for the verification comparisons. The adapter uses LoRA rank 8 and $\alpha = 16$. On a 400-example format audit, the model reaches 96.0% complete-section rate, 99.0% VGF parse success, 96.0% DSL syntax success, and 94.25% DSL execution success, comparable to the format reliability of the Qwen model.

I κ Matching Threshold Sensitivity

The relation $X_i \approx Y_i$ is implemented by normalizing the claimed final answer and executed DSL answer, then applying numeric tolerance. Floating-point answers, fractions, square roots, and π expressions are parsed to real values; degree symbols are treated as unit suffixes and removed before numeric comparison. For multiple-choice questions, numeric answers are first mapped to option letters using the fixed choice tolerance 0.02. Table I.1 reports an ablation that keeps the numeric absolute tolerance fixed at 10^{-3} and sweeps only the candidate-level relative tolerance used for κ.

All threshold settings yield identical results because geometry answers are effectively discrete: integers, simple fractions, radicals, angles, or multiple-choice letters. In this domain, claim–execution agreement is almost always binary: pairs either match exactly, or differ by margins much larger than even the loosest tolerance. The matching tolerance is therefore not a meaningful hyperparameter for these evaluation sets, and the measured CCV gain is not an artifact of finely tuned numeric equality.

Beyond the candidate-level relative tolerance, we swept every remaining fixed constant on both models. The absolute tolerance (default 10^{-3}) and the answer-clustering tolerance (default 0.005) leave CCV accuracy unchanged across four orders of magnitude. The multiple-choice mapping

Table I.1: Sensitivity to the candidate-level relative tolerance in $X_i \approx Y_i$ on the frozen Qwen3.5-9B K=16, T=0.8 rollout pool.

Setting	Candidate relative tolerance	CCV accuracy	κ candidates / precision
10^{-6}	0.000001	80.98	15,535 / 87.07
10^{-4}	0.0001	80.98	15,535 / 87.07
10^{-3}	0.001	80.98	15,535 / 87.07
default (10^{-2})	0.01	80.98	15,535 / 87.07
5×10^{-2}	0.05	80.98	15,535 / 87.07

tolerance (default 0.02) is stable until implausibly large values: 0.1 gives 73.14% on Qwen3.5-9B, while 0.001–0.01 give 81.28–81.33%, so the default is not locally optimal. Varying the claim weight (default 0.5) over $[0, 2]$ changes accuracy by at most 0.64 pp. The $+1$ bonus in Line 3 of Algorithm 1 cannot change the selection: every candidate in the κ-pool receives it, so it does not affect the argmax.

J Qualitative Case Studies

We present four representative cases from the frozen Qwen3.5-9B K=16, T=0.8 rollout pool, selected to illustrate the four behavioral modes of CCV: (1) successful rescue when execution consensus is wrong but claim is correct, (2) hard cases where both pathways agree on a wrong answer, (3) correlated perception errors that produce incorrect agreement, and (4) fallback routing when no self-consistent candidate exists. Each case reports method outputs and the key DSL of the CCV-selected candidate.

J.1 CCV Rescue (Exec✗ Claim✓)

`geometry3k_test` qid 325. Expected: B (16). Choices: A=12, B=16, C=51, D=65.

Method outputs. Claim MV: B; Exec MV: C; EAV: C; CCV: **B**.

CCV-selected DSL ($\kappa = 1$, claim cluster 9, exec cluster 6):

```
angle_X = angle_L
x = angle_X / 4
answer = x          # 16.25 -> B
```

CCV selects a κ-consistent candidate whose claim and execution agree on B, while EAV follows a wrong execution majority (C). This is the canonical recoverable failure: the neural pathway is correct but the symbolic pathway is corrupted by shared perception error.

J.2 Hard Case: Exec✗ Claim✗

`geometry3k_test` qid 72. Expected: A (3). Choices: A=3, B=4, C=6, D=7.

Method outputs. Claim MV: B; Exec MV: B; EAV: B; CCV: B.

CCV-selected DSL ($\kappa = 1$, claim cluster 15, exec cluster 15):

```
x = 4
answer = x          # 4.0 -> B
```

All methods agree on the wrong answer. This case sits outside selection-only verification because the cross-channel agreement is itself wrong—both pathways share the same perceptual error.

J.3 Correlated Perception Error

`geometry3k_test` qid 35. Expected: D (155). Choices: A=20, B=25, C=30, D=155.

Method outputs. Claim MV: B; Exec MV: B; EAV: B; CCV: B.

CCV-selected DSL ($\kappa = 1$, claim cluster 16, exec cluster 16):

```
arc_BC = 30
arc_AD = 20
angle_2 = (arc_BC + arc_AD) / 2
answer = angle_2    # 25.0 -> B
```

This is a boundary case for CCV: a correlated visual error makes both channels self-consistent around the wrong answer. With 16/16 candidates in both clusters, selection alone has no alternative evidence to break the incorrect consensus.

J.4 Empty κ-Pool with Execution Fallback

`geometry3k_test` qid 56. Expected: C (70). Choices: A=66, B=68, C=70, D=72.

Method outputs. Claim MV: D; Exec MV: C; EAV: C; CCV: **C**.

CCV-selected DSL ($\kappa = 0$, claim cluster 10, exec cluster 16):

```
leg1 = 24
hypotenuse = 74
x = sqrt(hypotenuse**2 - leg1**2)
answer = x          # 70.0 -> C
```

The κ-pool is empty (no candidate's claim matches its execution), so CCV falls back to the execution-majority pathway. The large exec cluster (16/16) exceeds τ_{exec}, providing the correct answer. This illustrates dual-pathway routing: when cross-channel agreement is absent, the deterministic symbolic path remains available as a fallback.

K Summary of Notation

Table K.1 summarizes the mathematical notations used throughout the paper.

Table K.1: Summary of notations and definitions.

Symbol	Definition
K	Number of sampled reasoning candidates per question (default $K = 16$)
z^*	Ground-truth correct answer
X_i	Natural-language claim (stated answer) of candidate i
Y_i	Deterministic execution result of the DSL program for candidate i
$\perp$	Value of a failed execution (e.g., syntax error, runtime exception)
κ_i	Indicator of cross-channel agreement: $\mathbf{1}[X_i \approx Y_i \wedge Y_i \neq \perp]$
p_e	Marginal execution accuracy: $P(Y_i = z^*)$ (failed executions count as incorrect)
p_κ	Precision of self-consistent candidates: $P(Y_i = z^* \mid \kappa_i = 1)$
a_1	Agreement rate given correct execution: $P(\kappa_i = 1 \mid Y_i = z^*)$
a_0	False agreement rate given incorrect execution: $P(\kappa_i = 1 \mid Y_i \neq z^*)$
t_κ	Joint probability of true agreement: $P(X_i = Y_i = z^*)$
f_κ	Joint probability of false agreement: $P(X_i = Y_i \neq z^*)$
ρ_e	Pairwise correlation of correctness between execution outputs of two candidates
ρ_κ	Pairwise correlation of correctness among retained κ-consistent candidates
K_κ	Number of self-consistent candidates retained in the κ-pool
ϕ	Correlation between claim correctness and execution correctness within a question
τ_{exec}	Execution-majority consensus threshold for fallback routing (default $\tau_{\text{exec}} = 9$)
N_{eff}	Effective sample size under correlated voters: $K/(1 + (K-1)\rho)$

L Proofs for the Cross-Channel Verification Analysis

L.1 Shared-perception model for execution

We formalize the execution verification problem using answer-valued random variables. Let Y_i denote the executed answer for candidate i, taking values in an answer space $\mathcal{Z}$ with correct answer z^*. A

latent visual state V is shared across candidates. Conditional on V, the executed answers Y_i are i.i.d. from $P_e(\cdot \mid V)$.

Define $p_e = P(Y_i = z^*) = \mathbb{E}_V[P_e(z^* \mid V)]$. For execution correctness indicators $E_i = \mathbf{1}[Y_i = z^*]$, the within-channel execution correlation is

$$\rho_e = \mathrm{Corr}(E_i, E_j) = \frac{\mathrm{Var}_V[P_e(z^* \mid V)]}{p_e(1 - p_e)}. \tag{L.1}$$

By the law of total covariance applied to $E_i = \mathbf{1}[Y_i = z^*]$ for $i \neq j$, $\mathrm{Cov}(E_i, E_j) = \mathrm{Var}_V[P_e(z^* \mid V)]$, so shared perception creates positive correlation whenever execution accuracy varies with the visual state.

L.2 Proof of Theorem 1

Let $\kappa_i = \mathbf{1}[X_i = Y_i]$. Decompose self-consistency into true and false agreement:

$$t_\kappa = P(X_i = Y_i = z^*), \tag{L.2}$$

$$f_\kappa = P(X_i = Y_i \neq z^*) = \sum_{z \neq z^*} P(X_i = z, Y_i = z). \tag{L.3}$$

Then $P(\kappa_i = 1) = t_\kappa + f_\kappa$, and by Bayes' rule:

$$P(Y_i = z^* \mid \kappa_i = 1) = \frac{t_\kappa}{t_\kappa + f_\kappa}. \tag{L.4}$$

This identity does not require independence; it simply states that self-consistent candidates are correct when agreement is true rather than false.

L.3 Proof of Proposition 1

By Bayes' rule, $p_\kappa = P(E_i = 1 \mid \kappa_i = 1) = \frac{a_1 p_e}{a_1 p_e + a_0(1-p_e)}$ and $1 - p_\kappa = \frac{a_0(1-p_e)}{a_1 p_e + a_0(1-p_e)}$. Dividing the two gives the odds identity. Odds are strictly increasing in probability, so $p_\kappa > p_e$ if and only if $a_1/a_0 > 1$. The derivation uses no assumption about how X_i and Y_i are generated.

L.4 Proof of Theorem 2

For n exchangeable binary correctness variables with $P(B_i = 1) = p$ and pairwise correlation ρ, the variance of the sample mean is $p(1-p)[1 + (n-1)\rho]/n$. Applying the SNR definition from the main text to raw exec voting (p_e, K, ρ_e) and to the retained κ-filtered pool $(p_\kappa, K_\kappa, \rho_\kappa)$ gives $\mathrm{SNR}_\kappa > \mathrm{SNR}_{\mathrm{raw}}$ iff

$$\frac{p_\kappa - \frac{1}{2}}{p_e - \frac{1}{2}} > \sqrt{\frac{p_\kappa(1-p_\kappa)}{p_e(1-p_e)} \cdot \frac{K(1 + (K_\kappa - 1)\rho_\kappa)}{K_\kappa(1 + (K-1)\rho_e)}}. \tag{L.5}$$

L.5 Numerical illustration

We instantiate the SNR proxy using measured post-hoc statistics from the full evaluation pool. Failed executions are counted as incorrect; correlations are Pearson correlations over ordered within-question candidate pairs. Raw exec voting uses $K = 16$, $p_e = 0.617$, and $\rho_e = 0.526$. The retained pool has mean size $K_\kappa = 9.09$, $p_\kappa = 0.871$, and $\rho_\kappa = 0.646$. Table L.1 summarizes the resulting SNR proxy values.

Table L.1: Measured statistics for the one-vs-rest SNR proxy.

Method	n	p	ρ	SNR
Raw exec MV	16	0.617	0.526	0.32
κ-filtered MV	9.09	0.871	0.646	1.33

The retained pool is therefore 23% more correlated, but the SNR proxy improves by $4.1\times$. This empirical regime is exactly the counterintuitive case captured by Theorem 2: precision filtering dominates the correlation penalty. On Qwen3.5-9B, 12.9% of κ=1 candidates are false agreements; on Gemma4-E4B the rate is 40.2%, and the precision gain persists, as Proposition 1 predicts.

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